

Option Contracts: Call and Put Payoff and Profit

LEARNING OBJECTIVES

By the end of this lecture, students will be able to:

- **Read an option contract.** Identify the underlying, strike, expiration, exercise style, premium, position direction, and contract multiplier.
- **Separate payoff from profit.** Derive long and short call and put expiration payoffs, premium-adjusted profit, breakeven prices, and loss bounds.
- **Compose option positions.** Represent a multi-leg strategy as a signed sum of contract cash flows without hiding units or financing conventions.

1 A Derivative Is a Contract on Another Quantity

A derivative is a financial contract whose cash flows depend on an underlying quantity such as an equity price, interest rate, currency, commodity, index, or credit event. Forwards and futures obligate counterparties to transact later; swaps exchange specified cash-flow streams; options give the holder a right while imposing a corresponding obligation on the writer if the right is exercised.

An option description is incomplete without its terms. Let $S_t \geq 0$ be the underlying price in USD/share at time t , let $K > 0$ be the strike in USD/share, and let $T > 0$ be expiration. A *call* gives its long holder the right to buy the underlying at K ; a *put* gives its long holder the right to sell at K . European-style exercise occurs only at T , while American-style exercise may occur on or before T . This lecture analyzes expiration cash flows; early-exercise decisions require a time-dependent valuation model.

The premium is paid when the position opens. Let $C_0 \geq 0$ and $P_0 \geq 0$ be call and put premiums in USD/share at time 0. Let $q \in \mathbb{N}$ be the deliverable multiplier in shares/contract. Standard U.S. equity options commonly use $q = 100$, but adjusted and nonstandard contracts can differ; the actual contract specification controls.

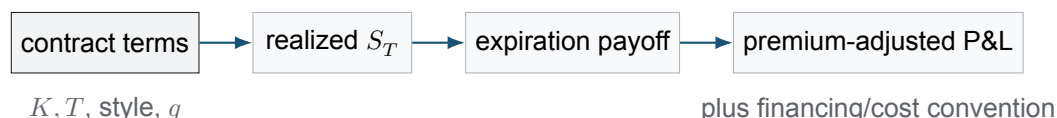


Figure 1: Payoff is fixed by the contract and the terminal underlying price. Profit additionally depends on the opening premium, position direction, multiplier, financing convention, fees, and any intervening exercise or assignment.

2 Expiration Payoffs Are Contractual

For any real x , write $x^+ := \max(x, 0)$. The per-share terminal payoff to a long European call and long European put is

$$h_c(S_T; K) := (S_T - K)^+, \quad h_p(S_T; K) := (K - S_T)^+. \quad (1)$$

The call is exercised at expiration when buying at K is better than buying at S_T ; the put is exercised when selling at K is better than selling at S_T . The holder may decline exercise, so both payoffs are nonnegative.

Let $\theta \in \{-1, +1\}$ denote position direction, with $+1$ long and -1 short. Ignoring counterparty default, the writer's terminal payoff is the negative of the holder's payoff:

$$V_T^{c,\theta} = \theta h_c(S_T; K), \quad V_T^{p,\theta} = \theta h_p(S_T; K). \quad (2)$$

This payoff symmetry does not imply symmetric risk after the premium: call and put terminal functions have different bounds because $S_T \geq 0$ but has no finite contractual upper bound.

Definition 2.1. Moneyness at expiration

A call is in the money when $S_T > K$, at the money when $S_T = K$, and out of the money when $S_T < K$. A put reverses the inequalities. Moneyness describes exercise value relative to the strike; it does not by itself determine whether the original position earned a profit because the premium was paid or received separately.

3 Profit Requires a Cash-Flow Convention

The common classroom expiration diagram subtracts the opening premium without growing it through time. Under this undiscounted cash-flow convention, per-share profit is

$$\Pi_T^{c,\theta}(S_T) = \theta(h_c(S_T; K) - C_0), \quad \Pi_T^{p,\theta}(S_T) = \theta(h_p(S_T; K) - P_0). \quad (3)$$

For a long position, the premium is an initial outflow; for a short position, it is an initial inflow. Equation (3) is useful for shape and breakeven calculations, but it mixes time-0 and time- T dollars. A time- T economic P&L would subtract the premium's financed future value, while a time-0 net present value would discount the terminal payoff. Fees and bid-ask effects must also be added consistently.

Under the classroom convention, the long call breakeven solves $S_T - K - C_0 = 0$ in the exercised region, and the long put breakeven solves $K - S_T - P_0 = 0$. Therefore

$$S_{\text{BE}}^c = K + C_0, \quad S_{\text{BE}}^p = K - P_0. \quad (4)$$

The same prices are breakevens for the corresponding short positions because their profits are exact negatives under Eq. 3. If $P_0 > K$, the algebraic put breakeven is negative and therefore outside the feasible domain $S_T \geq 0$.

4 Loss Bounds and Direction Matter

For one share-equivalent under the convention in Eq. 3, the terminal bounds are summarized below. Multiply every amount by the deliverable q and by the number of contracts to obtain position-level dollars.

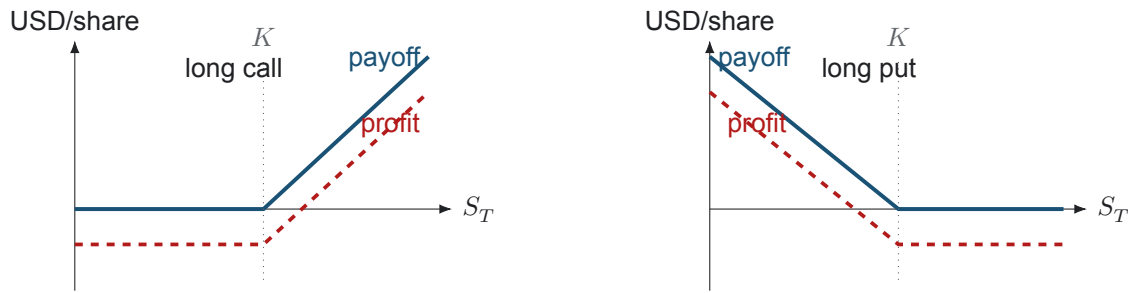


Figure 2: Long call and put payoffs (solid blue) and classroom profit diagrams (dashed red). The premium shifts profit downward without changing the kink at K ; breakeven is not the same point as the strike.

Position	Maximum profit	Maximum loss	Breakeven
Long call	unbounded as $S_T \rightarrow \infty$	C_0	$K + C_0$
Short call	C_0	unbounded as $S_T \rightarrow \infty$	$K + C_0$
Long put	$K - P_0$ at $S_T = 0$	P_0	$K - P_0$
Short put	P_0	$K - P_0$ at $S_T = 0$	$K - P_0$

The short put's terminal loss is large but finite when the underlying price is bounded below by zero. Calling it “unlimited downside loss” is incorrect for an ordinary put with finite strike. By contrast, an uncovered short call has no finite contractual loss bound because the underlying price has no finite upper bound. Margin requirements, early assignment, liquidity, and forced liquidation can make realized risk severe well before terminal bounds are reached.

COMPANION NOTEBOOK

L8b: Single-Contract Payoff and Profit →

Build all four positions from an option chain, sweep the full domain $S_T \geq 0$, convert per-share quantities to per-contract dollars, and verify the breakevens and both loss bounds numerically.

5 Composite Positions Add Leg by Leg

Let a composite position $\mathcal{C}$ contain legs $i \in \{1, \dots, d\}$ on a common underlying and expiration. For leg i , let $\tau_i \in \{c, p\}$ be the type, $K_i > 0$ the strike, $\theta_i \in \{-1, +1\}$ the direction, $n_i \in \mathbb{N}$ the number of contracts, and $P_i \geq 0$ the per-share opening premium. Write the long per-share payoff of leg i as

$$h_i(S_T) := \begin{cases} (S_T - K_i)^+, & \tau_i = c, \\ (K_i - S_T)^+, & \tau_i = p. \end{cases} \quad (5)$$

Under the same undiscounted convention, the terminal payoff and profit of the position, *per share*, are

$$H_{\mathcal{C}}(S_T) = \sum_{i=1}^d \theta_i n_i h_i(S_T), \quad (6)$$

$$\Pi_{\mathcal{C}}(S_T) = \sum_{i=1}^d \theta_i n_i [h_i(S_T) - P_i]. \quad (7)$$

Both are in USD/share, which is what the example notebooks compute and plot. To convert to position-level dollars, multiply leg i by its deliverable $q_i \in \mathbb{N}$ in shares/contract: the position payoff is $\sum_i \theta_i n_i q_i h_i(S_T)$, and

standard U.S. equity options carry $q_i = 100$. Keeping q_i outside the per-share sum matters because a corporate-action adjustment can change one leg's deliverable while leaving the others standard.

Linearity makes vertical spreads, straddles, and strangles transparent: corners can occur only at component strikes, and the vertical offset depends on the net opening debit or credit. A strike is an actual corner only when the aggregate slope jump there, $\sum_{i:K_i=K} \theta_i n_i$, is nonzero; opposing legs at one strike cancel. Covered calls and protective puts need one further building block, a position in the underlying shares, because Eq. 5 admits only $\tau_i \in \{c, p\}$. Exercise style, expiration, underlying, deliverable, and settlement must be compatible before legs are casually combined into one diagram.

The domain constrains composite bounds exactly as it does single contracts. Writing D for the net per-share debit of a long straddle at strike K , its profit $|S_T - K| - D$ is unbounded above only as $S_T \rightarrow \infty$; on the lower branch it terminates at $S_T = 0$ with the finite value $K - D$. The familiar description “unlimited profit in either direction” is therefore wrong on the downside, and the corresponding short position's downside loss is likewise finite.

Remark 5.1. Which payoffs are reachable

Equation (6) shows that summing legs *produces* piecewise-linear payoffs. The converse also holds: for a fixed strike grid $0 < K_1 < \dots < K_d$, every continuous piecewise-linear payoff whose corners lie on that grid has exactly one representation as cash, shares of the underlying, and calls struck at those strikes, with each call weight equal to the slope jump at its strike. The minimal representation uses exactly the strikes whose slope jump is nonzero. The optional advanced notebook proves it and reads the standard structures off the decomposition.

COMPANION NOTEBOOK

L8b: Composite-Contract Payoff and Profit →

Combine signed option legs to study vertical spreads, straddles, and strangles, and identify their piecewise profit regions, breakevens, and finite downside extremes at $S_T = 0$.

COMPANION NOTEBOOK

L8b optional advanced: Static Replication of Piecewise-Linear Payoffs →

Proves Remark 5.1: existence and uniqueness of the cash, share, and call decomposition, the put payoff identity, the decomposition of every strategy above, and the boundary of the reachable set. Optional; not a prerequisite for L9a.

KEY TAKEAWAYS

In this lecture, we translated option contract terms into expiration cash flows and kept contractual payoff distinct from the investor's premium-dependent profit.

- **Payoff belongs to the contract.** A long call pays $(S_T - K)^+$ and a long put pays $(K - S_T)^+$; short positions reverse those cash flows.
- **Profit needs a convention.** Premium, financing time, multiplier, fees, and position direction determine P&L, so strike, moneyness, and breakeven are different concepts.
- **Loss bounds are asymmetric.** An uncovered short call has unbounded terminal loss, while a standard short put's maximum terminal loss is finite at $S_T = 0$.

Next, value European claims with Black–Scholes–Merton and American claims with a Cox–Ross–

Rubinstein exercise lattice.**ADDITIONAL READING**

- John C. Hull, *Options, Futures, and Other Derivatives*, 9th ed., Pearson (2015), chapters on option-market mechanics and payoff strategies.
- Options Clearing Corporation, *Characteristics and Risks of Standardized Options*, for contract, exercise, assignment, and risk disclosures.
- Robert C. Merton, “Theory of Rational Option Pricing,” *Bell Journal of Economics and Management Science* 4(1), 141–183 (1973).

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